



JKBOSE 2026

CLASS 11TH

CONIC SECTIONS

MATHS

LECTURE-07

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TOPICS TO BE COVERED

1

Hyperbola

2

Related Problems

Hyperbola

The locus of a point $P(x, y)$ in a plane such that the difference of its distance from two fixed points is always constant, is known as the hyperbola.

OR

A hyperbola is a set of points in a plane, difference of whose distances from two fixed points in a plane, is always constant.

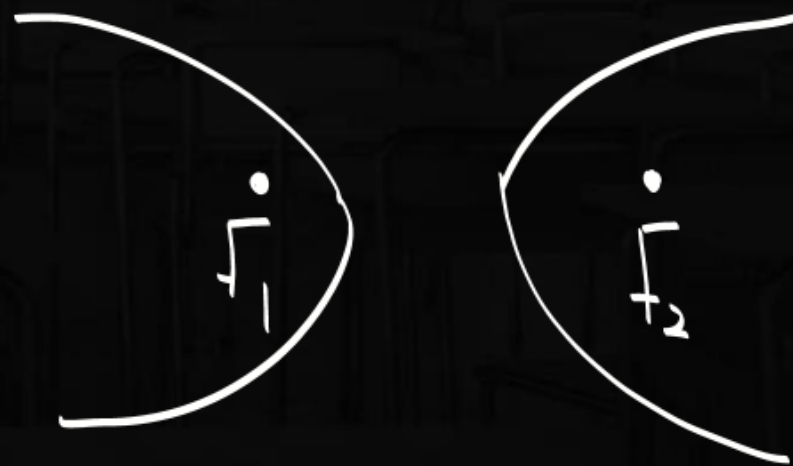
Suppose that $P(x, y)$ be any point on the hyperbola

& F_1, F_2 are the two given fixed points, then

$$|PF_1 - PF_2| = \text{Constant.}$$



* Fixed points are known as the foci of the hyperbola



★ Terms Related To Hyperbola:

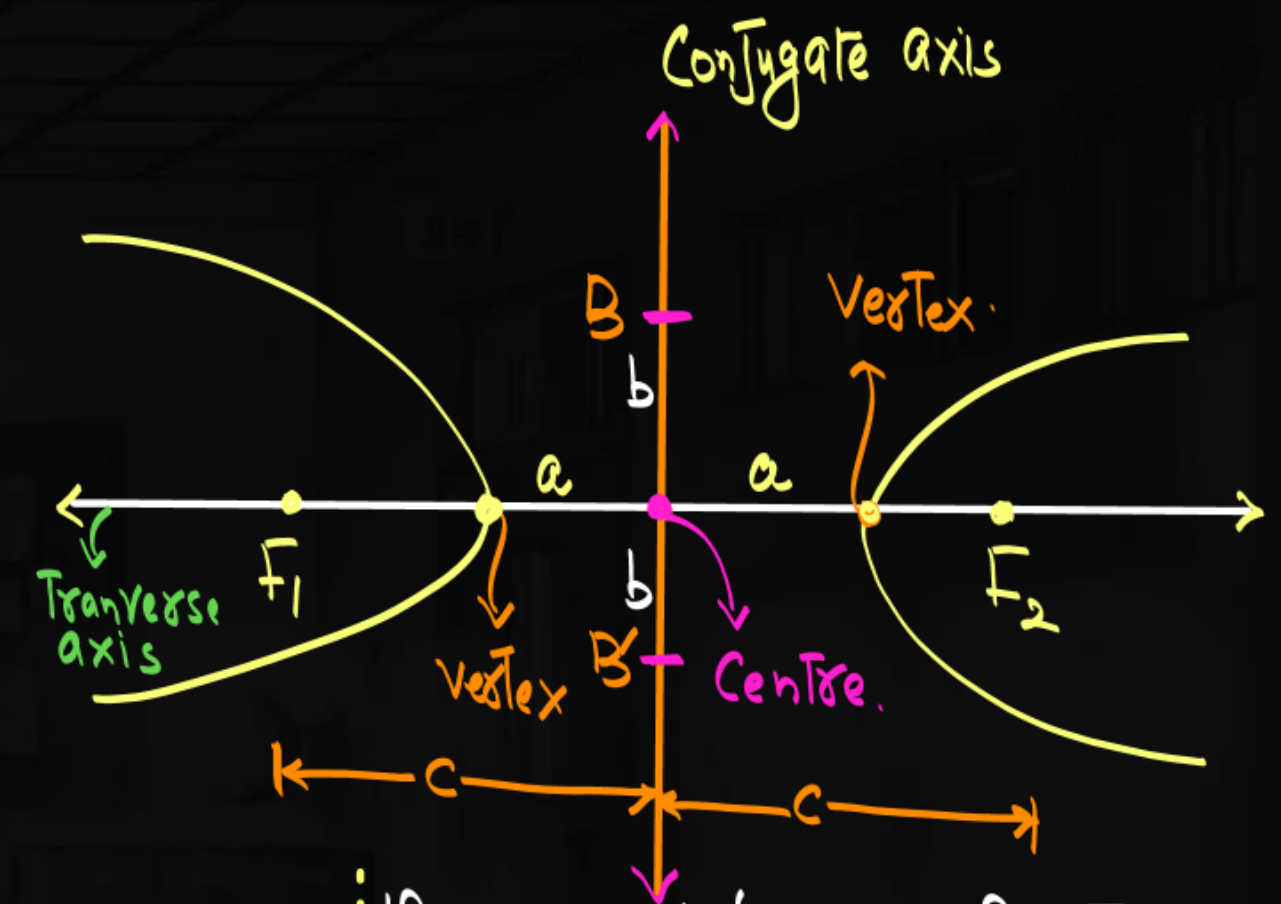
Transverse axis Line passing through the foci of the Hyperbola is known as Transverse axis.

Centre of the Hyperbola The mid point of the line segment joining the two foci is known as the Centre of the Hyperbola.

Vertices of the Hyperbola Points of Intersection of the Hyperbola and the Transverse axis are known as vertices of the Hyperbola.

Conjugate axis Line perpendicular to the Transverse axis and passing through the Centre of the Hyperbola is known as Conjugate axis.

- * a → Distance of one of the vertex from the Centre.
- * b → Distance of one of the points on the Conjugate axis from the Centre.
- * c → Distance of one of the foci from the Centre.



Here, we define the Quantity b such that the following relation is satisfied:

$$b = \sqrt{c^2 - a^2}$$

$$\Rightarrow b^2 = c^2 - a^2$$

$$\Rightarrow c^2 = a^2 + b^2$$

$$\Rightarrow c = \sqrt{a^2 + b^2}$$

* $2a$ $\longrightarrow$ Distance b/w the vertices of the hyperbola (length of Transverse axis)

* $2b$ $\longrightarrow$ Distance b/w B & B' (length of conjugate axis)

* $2c$ $\longrightarrow$ Distance b/w the two foci of the hyperbola.

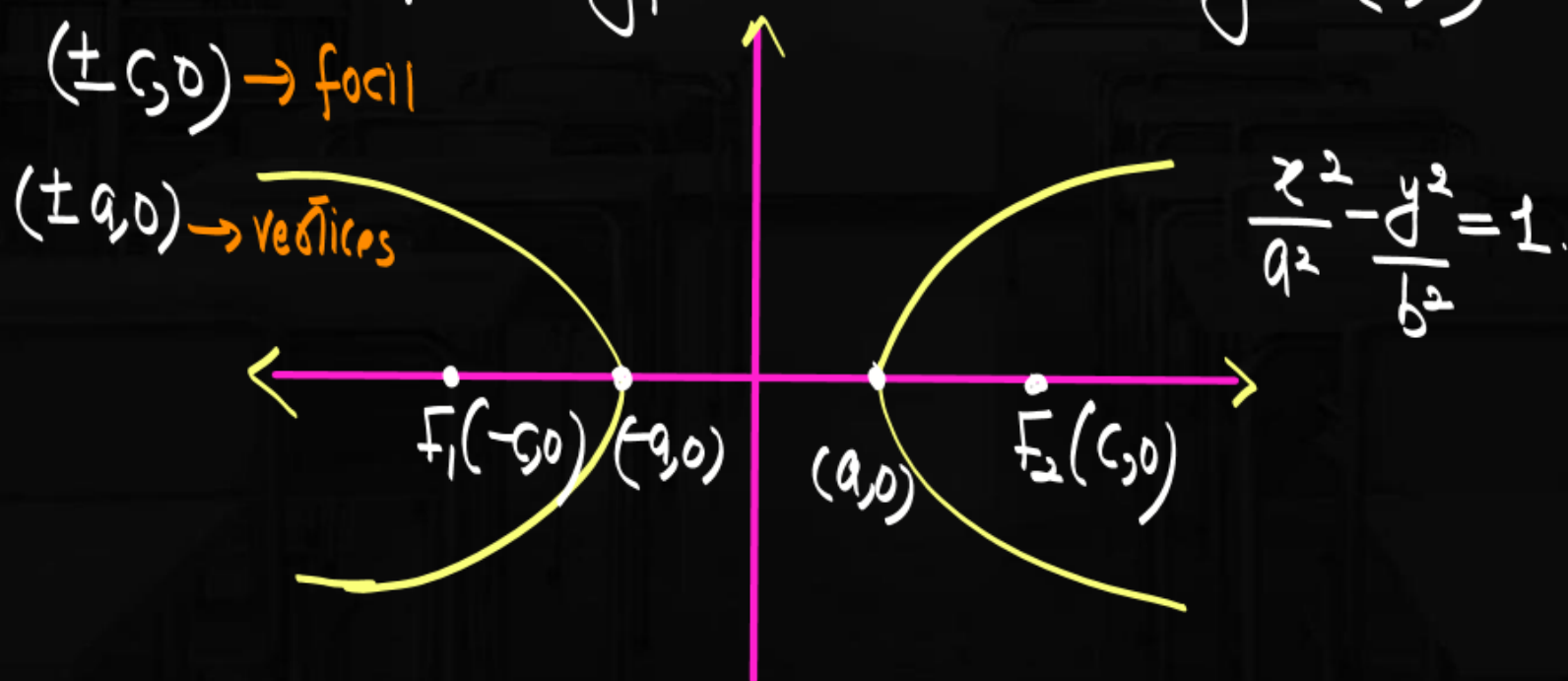
Equation of the hyperbola in standard form:

* Transverse axis & conjugate axis are along co-ordinate axis.

* Centre of the hyperbola is at the origin $(0,0)$.

$(\pm c, 0) \rightarrow$ foci

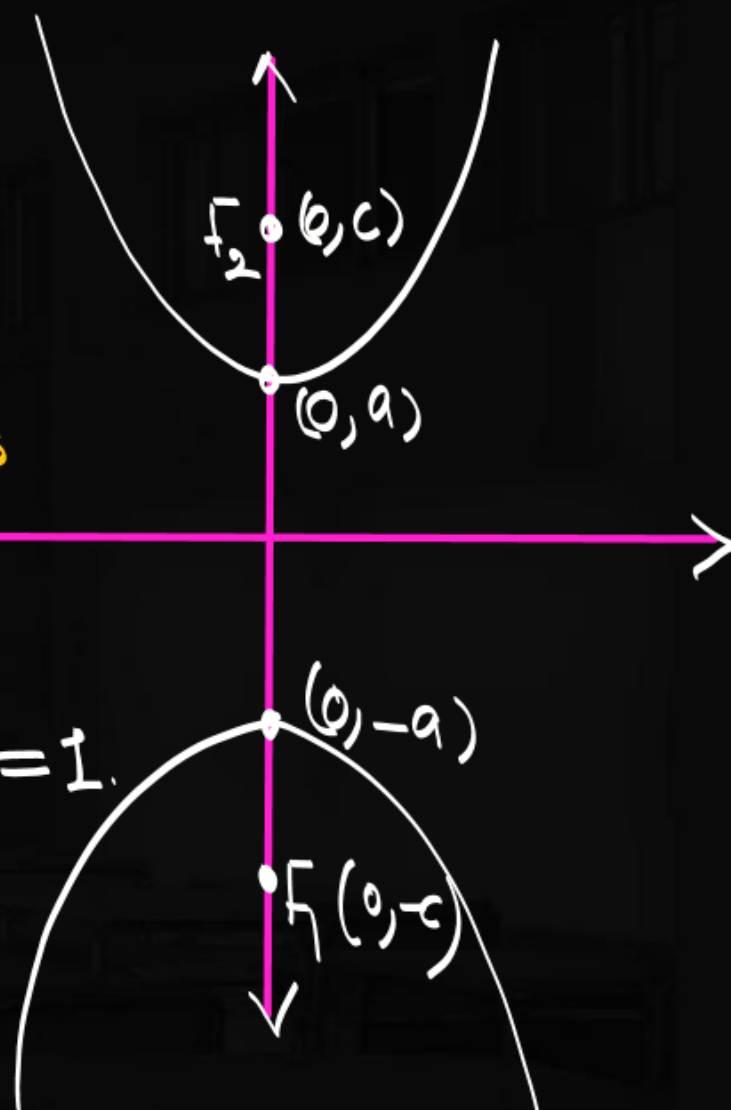
$(\pm a, 0) \rightarrow$ vertices



$(0, \pm c) \rightarrow$ foci

$(0, \pm a) \rightarrow$ vertices

$$\frac{y^2}{a^2} - \frac{x^2}{b^2} = 1$$



latus Rectum:

Chord passing through any focus & perpendicular to the Transverse axis is known as latus Rectum

* length of latus Rectum = $\frac{2b^2}{a}$

eccentricity of the hyperbola: The ratio of the distance of one of the foci from the centre & the distance of one of the vertex from the centre is known as eccentricity of the hyperbola.

* Denoted by e . So, $e = \frac{c}{a}$.

$$\text{Here } c > a \Rightarrow \frac{c}{a} > 1 \Rightarrow e > 1$$

$\Rightarrow$ eccentricity of the hyperbola is greater than 1.

$$\text{Hso, } e^2 = \frac{c^2}{a^2} = \frac{a^2 + b^2}{a^2} = \frac{a^2}{a^2} + \frac{b^2}{a^2}$$

$$\Rightarrow e^2 - 1 = \frac{b^2}{a^2}$$

$$\Rightarrow b^2 = a^2(e^2 - 1)$$

★ **Rectangular Hyperbola / Equilateral Hyperbola:**

Hyperbola in which the length of Transverse axis is equal to the length of Conjugate axis is known as the equilateral hyperbola.

$$\text{Here } e = \frac{c}{a} = \frac{\sqrt{a^2 + b^2}}{a} = \frac{\sqrt{a^2 + a^2}}{a} = \frac{\sqrt{2a^2}}{a}$$

$$\Rightarrow e = \frac{\sqrt{2}a}{a} \Rightarrow e = \sqrt{2}$$

$\Rightarrow$ eccentricity of the equilateral hyperbola is $\sqrt{2}$.

EXAMPLE

Find the coordinates of the **foci** and the **vertices**, the **eccentricity**, the **length** of the latus rectum of the hyperbolas:

(i) $\frac{x^2}{9} - \frac{y^2}{16} = 1$

(ii) $y^2 - 16x^2 = 16$

(Sol) (i) Given hyperbola is:

$$\frac{x^2}{9} - \frac{y^2}{16} = 1$$

$$\Rightarrow \frac{x^2}{3^2} - \frac{y^2}{4^2} = 1$$

$$\Rightarrow a = 3, b = 4$$

$$\text{So, } c = \sqrt{a^2 + b^2} = \sqrt{9 + 16}$$

$$\Rightarrow c = 5$$

$$\text{Therefore foci} = (\pm 5, 0)$$

$$\text{Vertices} = (\pm 3, 0)$$

$$\text{eccentricity, } e = \frac{c}{a} = \frac{5}{3}$$

$$\begin{aligned} \text{length of latus Rectum} &= \frac{2b^2}{a} \\ &= \frac{2 \times 16}{3} = \frac{32}{3} \end{aligned}$$

(ii) Given hyperbola is:

$$y^2 - 16x^2 = 16$$

$$\Rightarrow \frac{y^2}{16} - \frac{x^2}{1} = 1$$

$$\Rightarrow \frac{y^2}{4^2} - \frac{x^2}{1^2} = 1$$

$$\Rightarrow a = 4, b = 1$$

$$\text{So, } c = \sqrt{a^2 + b^2} = \sqrt{17}$$

$$\text{Therefore foci} = (0, \pm \sqrt{17})$$

$$\text{Vertices} = (0, \pm 4)$$

$$\text{eccentricity, } e = \frac{c}{a} = \frac{\sqrt{17}}{4}$$

length of latus Rectum

$$= \frac{2b^2}{a}$$

$$= \frac{2 \times 1^2}{4}$$

$$= \frac{1}{2}$$

Hence the result.

EXAMPLE

Find the equation of the hyperbola where foci are $(0, \pm 12)$ and the length of the latus rectum is 36.

(Sol) It's given that $c = 12$ & $\frac{2b^2}{a} = 36$

$$\Rightarrow c^2 = 144 \text{ & } b^2 = 18a \text{ --- (1)}$$

$$\Rightarrow a^2 + b^2 = 144 \text{ --- (2)}$$

Using (1) in (2), we get;

$$a^2 + 18a = 144$$

$$\Rightarrow a^2 + 18a - 144 = 0$$

$$\Rightarrow a^2 + 24a - 6a - 144 = 0$$

$$\Rightarrow a(a+24) - 6(a+24) = 0$$

$$\Rightarrow (a-6)(a+24) = 0$$

$$\Rightarrow a-6 = 0$$

$$\Rightarrow a = 6 \Rightarrow a^2 = 36$$

So from (1), we get;

$$b^2 = 18 \times 6 = 108$$

$$\Rightarrow b^2 = 108$$

Therefore the required hyperbola is:

$$\frac{y^2}{a^2} - \frac{x^2}{b^2} = 1$$

$$\Rightarrow \frac{y^2}{36} - \frac{x^2}{108} = 1$$

$$\Rightarrow 3y^2 - x^2 = 108$$

Hence the result.

QUESTION (NCERT EXAMPLER):

The eccentricity of the hyperbola whose latus rectum is 8 and conjugate axis is equal to half of the distance between the foci is.

1 $\frac{4}{3}$ ✗

2 $\frac{4}{\sqrt{3}}$ ✗

3 $\frac{2}{\sqrt{3}}$ ✓✓

4 None of these

(Sol) It's Given that $\frac{2b^2}{a} = 8$.

$\Rightarrow b^2 = 4a$ — ①

Also $2b = \frac{2c}{2} = c$.

$\Rightarrow 2b = c$

$\Rightarrow (2b)^2 = c^2$

$\Rightarrow 4b^2 = a^2 + b^2$

$\Rightarrow 3b^2 = a^2$

$\Rightarrow a = \sqrt{3}b$

So, $e = \frac{c}{a}$.

$\Rightarrow e = \frac{2b}{\sqrt{3}b}$

$\Rightarrow e = \frac{2}{\sqrt{3}}$

QUESTION (NCERT EXAMPLER):

The distance between the foci of a hyperbola is 16 and its eccentricity is $\sqrt{2}$. Its equation is

1 $x^2 - y^2 = 32$ *(Homework Problem)*

2 $\frac{x^2}{4} - \frac{y^2}{9} = 1$

3 $2x - 3y^2 = 7$

4 None of these

QUESTION (NCERT EXAMPLER):

Equation of the hyperbola with eccentricity $\frac{3}{2}$ and foci at $(\pm 2, 0)$ is.

1 $\frac{x^2}{4} - \frac{y^2}{5} = \frac{4}{9}$ (Homework Problem)

2 $\frac{x^2}{9} - \frac{y^2}{9} = \frac{4}{9}$

3 $\frac{x^2}{4} - \frac{y^2}{9} = 1$

4 None of these

QUESTION (JEE-MAINS 2016)

The **eccentricity** of the **Hyperbola** whose length of the **Latus rectum** is equal to 8 and the length of its **conjugate axis** is equal to half of the distance between the Foci, is:

1 $\sqrt{3}$ ✗

2 $\frac{4}{3}$ ✗

3 $\frac{4}{\sqrt{3}}$ ✗

4 $\frac{2}{\sqrt{3}}$ ✓✓

Main Facts about the Hyperbola

Forms of the Hyperbola	$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$	$\frac{y^2}{a^2} - \frac{x^2}{b^2} = 1$
Equation of transverse axis	$y = 0$	$x = 0$
Equation of conjugate axis	$x = 0$	$y = 0$
Length of transverse axis	$2a$	$2a$
Foci	$(\pm ae, 0)$	$(0, \pm ae)$
Equation of latus rectum	$x = \pm ae$	$y = \pm ae$
Length of latus rectum	$\frac{2b^2}{a}$	$\frac{2b^2}{a}$
Centre	$(0, 0)$	$(0, 0)$

Tamana 

Rasib Ayoub Rather 

Para special forces. 

Thank
You